

Joshua Lim

Ph.D. Student at Duke University, Department of Statistical Science

Durham, NC • josh.lim@duke.edu

EDUCATION

Duke University

Ph.D. Statistical Science

M.S. Computer Science

Certificate in College Teaching

Durham, NC

May 2029

University of California, Los Angeles (UCLA)

B.S. Applied Mathematics, Statistics and Data Science

- Summa Cum Laude, GPA: 4.0

Los Angeles, CA

June 2024

RESEARCH INTERESTS

Causal Inference, Machine Learning, Biostatistics, Missing Data, Monte Carlo Methods

RESEARCH EXPERIENCE

Duke University - Department of Statistical Science

Research Assistant

Durham, NC

May 2025 - Present

- Developing and implementing causal inference methodology and applications under the guidance of Professor Jerry Reiter and Fan Li
- Integrating LLMs and AI to enhance causal discovery

Duke University School of Medicine

Biostatistician

Durham, NC

May 2025 - Present

- Implemented statistical analysis to investigate the relative impact of contrast density and timing of image acquisition on CVF visibility
- Utilized techniques such as ordinal regression to provide analyses to radiologists
- Authored abstract, presentation, and papers submission to multiple neuroradiology conferences

UC San Diego School of Medicine - Moores Cancer Center

Biostatistician

Durham, NC

December 2025 - Present

- Implemented statistical modeling to compare treatment-related complications between patients receiving immunotherapy with radiation (IRT) versus concurrent chemoradiation with cisplatin

Stanford School of Medicine - Pulmonary, Allergy and Critical Care Medicine

Biostatistician and Data Manager

Stanford, CA

June 2024 - March 2025

- Implemented statistical analysis in R to identify environmental factors corresponding to increased asthma and COPD ER visits and hospitalizations
- Tracked regional and county-wide differences, paying special attention to vulnerable populations

- Second authored abstract and presentation submission to the 2025 American Academy of Allergy, Asthma & Immunology conference

Qing Zhou Research Group

Undergraduate Researcher

Los Angeles, CA

February 2023 - June 2024

- Researched structure learning of causal DAGs under the guidance of Professor and Department Chair Qing Zhou
- Furthered the development of the CML algorithm, created by Ph.D. student Stephen Smith, by adding in conditional independence tests and other procedures for discrete data
- Verified algorithm by generating datasets from the parameters provided by the Bnlearn repository and performing simulation studies on the Hoffman2 Cluster
- Implementation done in Rcpp, Rcpp Armadillo, and R

Stanford School of Medicine - Sean N. Parker Center for Allergy and Asthma Research

Stanford, CA

Biostatistics Researcher

June 2022 - May 2023

- Implemented statistical analysis in R to analyze patients with chronic GI symptoms in oral immunotherapy clinical trials
- First authored abstract and poster submission to the 2023 American Academy of Allergy, Asthma & Immunology conference and second authored poster submission to GALEN Anacare GAFA
- Generated machine learning models to automatically convert CTCAE, Bock, and CoFAR 1 allergic reaction grades to a CoFAR 3 grade
- Developed interactive R shiny application displaying patient enrollment, demographic, and forecast data. Used by researchers to provide summaries to the NIH

PUBLICATIONS

- Tirumalasetty, J., **Lim, J.**, Eggert, L., Arroyo, A., & Nassikas, N. (2025) “Associations of Air Pollution and Heat with Asthma Healthcare Visits in California Counties – An Ecological Study.” *Journal of Allergy and Clinical Immunology*, vol. 155, no. 2 (Supplement), AB309, <https://doi.org/10.1016/j.jaci.2024.12.309>
- Zhu, D., Kranz, P. G., Edelmut, D. G. L., **Lim, J.**, Christensen, S., Madhavan, A. A., & Amrhein, T. J. (2025) “Maximizing the Conspicuity of CSF–Venous Fistulas on CT Myelography: Assessment of Contrast Density and Timing Effects.” *American Journal of Neuroradiology*, <https://doi.org/10.3174/ajnr.A8972>.
- **Lim, J.**, Cao S., Chinthrajah S., Sindher S. (2023) “The Impact of Persistent Gastrointestinal Symptoms on Completion of Oral Immunotherapy in Clinical Trials.” *Journal of Allergy and Clinical Immunology*, vol. 151, no. 2, <https://doi.org/10.1016/j.jaci.2022.12.110>.

POSTERS AND PRESENTATIONS

- **Maximizing the Conspicuity of CSF-Venous Fistulas on CT Myelography: Assessment of Contrast Density and Timing Effects** November-December 2025
Radiological Society of North America Annual Meeting

- Enhancing Conspicuity of CSF-Venous Fistulas on CT Myelography: Assessment of Contrast Density and Timing Effects**
2025 Intracranial Hypotension Conference
June 2025
- Associations Of Air Pollution And Heat With Asthma Healthcare Visits In California Counties – An Ecological Study**
2025 AAAAI/WAO Joint Congress
March 2025
- Persistent gastrointestinal symptoms do not impact oral immunotherapy completion**
2023 AAAAI Annual Meeting
February 2023
- The impact of persistent gastrointestinal symptoms on completion of oral immunotherapy in clinical trials**
GALEN Anacare GAFA 2022
September 2022

TEACHING EXPERIENCE

Duke University - Department of Statistical Science

Durham, NC

Instructor of Record

June 2026 - August 2026

- STA 199: Introduction to Data Science and Statistical Thinking
- Preparing to teach a core statistics course for official Duke University Course credit, replete with hand-crafted lectures, labs, homework, and exams.

University of California, Los Angeles (UCLA) - Department of Statistics and Data Science

Los Angeles, CA

Teaching Assistant

January 2024 - June 2024

- Stats 13: Introduction to Statistical Methods for Life and Health Sciences
- Stats 102B: Computation and Optimization for Statistics
- Created original homework assignments and practice exams, led office hours and review sessions, and graded for sections of over 180 students.

Math Speeder Upper Course (MSUC)

Instructor and Curriculum Designer

June 2021 - August 2021

- Created a fully online introductory multivariable calculus course, featuring 22 online lectures, homework assignments, solutions, as well as a handful of quizzes, two midterms, three projects, and a final exam
- Welcomed 10 students in the first cohort, with several new students referencing the now fully asynchronous class every year
- Intended to be accessible by any student, regardless of background or socioeconomic status

Menlo School

Atherton, CA

Teaching Assistant

August 2020 - June 2021

- Assisted in the online, hybrid, and in-person learning formats for AP Statistics and AP Calculus BC
- Created test questions, offered help during and after school hours, and led classroom activities
- Increased classroom productivity by facilitating problem solving sessions with immediate feedback

Physics Catcher Upper Course

Instructor and Curriculum Designer

June 2020 - August 2020

- Created a fully online introductory physics course, featuring 17 online lectures, homework assignments, and answer keys, as well as a midterm and final exam

Tutor

June 2019 - Present

- Taught dozens of middle school to college-age students for over seven years in mathematics, statistics, and physics

LEADERSHIP AND SERVICE

Introduction to Undergraduate Research Workshop Mentor

May 2025

Duke University Department of Statistical Science

- Mentored a team of six undergraduate students through a comprehensive applied statistical analysis on the perceived quality of life in Durham, NC
- Executed mini workshops on good research practices, coding practices, and effective collaboration for all 20+ participants in the program
- Completed additional mentoring training under the guidance of Dr. Joan Combs Durso, coordinator of undergraduate training, research, and development

Graduate Consultative Committee (GCC)

August 2024 - Present

Duke University Department of Statistical Science

- Collaborated with the director of graduate studies and chair to manage academic and personal needs of >50 PhD students.
- Planned and executed admitted students open house events to welcome prospective students.
- Organized department social and community events such as the Halloween party, winter party, and intramural volleyball team

UCLA Club Gymnastics President

August 2022 - Present

- Served as the president for a nationally-ranked and recognized club gymnastics team of over 60 members
- Organized meetings, UCLA competitions, and travel meets, communicated with governing bodies, registered for competitions, fundraised, and more

TECHNICAL SKILLS

- R, Python, C++, MATLAB, Java, Git, Rcpp, Cloud Computing

AWARDS

- **Dean's Graduate Fellowship**

August 2024

Awarded to incoming Duke University PhD students who demonstrate a track record of excellent leadership

- **Summa Cum Laude**

June 2024

Awarded to UCLA students earning above a 3.985 cumulative GPA

- **Dean's Honor List (Fall 2020 - Spring 2024)**

December 2020 - Present

Awarded to UCLA students achieving a 3.9 GPA or above on at least 12 units

REFERENCES

Jerry Reiter, Ph.D.

Professor, Department of Statistical Science
Duke University
jreiter@duke.edu

Fan Li, Ph.D.

Professor, Department of Statistical Science
Duke University
fl35@duke.edu

Qing Zhou, Ph.D.

Professor and Chair, Statistics and Data Science Department
University of California, Los Angeles
zhou@stat.ucla.edu

Jyothi Tirumalasetty, MD

Clinical Assistant Professor, Medicine
Stanford University
jtsetty@stanford.edu

Michael Tsiang, Ph.D.

Continuing Lecturer, Statistics and Data Science Department
University of California, Los Angeles
michael.tsiang@stat.ucla.edu
